Syllabus
Semester-II

DATA STRUCTURES
Syllabus

Prerequisite: C programming and Basic of Mathematics.
L-T-P: 2-0-2, Credits: 3 Type: Engineering Essential/ Core Essential

Course Description

This course introduces the student's fundamentals of data structures and takes them forward to software design along with the course on Algorithms. It details how the choice of data structures impacts the performance of programs for given software application. This is a precursor to DBMS and Operating Systems. A lab course is associated with it to strengthen the concepts.

Course Objectives

Students undergoing this course are expected to:	
1	Give Knowledge of elementary Data organization, Complexity, Revision of Programming concepts
2	Give Basic& Advanced Knowledge of Array, Stack ,Queue& Linked List
3	Give Basic & Advanced Knowledge of Tree
4	Give Basic & Advanced Knowledge of Searching and Sorting
5	Give Basic & Advanced Knowledge of Tree Graph
6	Give Knowledge of elementary Data organization, Complexity, Revision of Programming concepts

Course Outcomes

CO Numbers	Course Outcomes (Action verb should be in italics	Bloom's taxonomy
CO-1	Students Understood elementary Data organization, Complexity, Revision of Programming concepts	Identify/Knowledge
CO-2	Students Understood Basic &Advanced Knowledge of Array, Stack ,Queue& Linked List	Understanding /Analyzing
CO-3	Students Understood Basic Advanced Knowledge of Tree	Understanding
CO-4	Students Understood Basic& Advanced Knowledge of Searching and Sorting	Applying
CO-5	Students Understood Basic& Advanced Knowledge of Tree Graph	Understanding/Comprehension

Mapping Course Outcomes with Program Outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10
CO1	3	2	3	2	2	0	0	1	2	0
CO2	2	2	3	3	1	0	1	1	2	1
CO3	1	3	3	2	2	0	2	2	1	1
CO4	2	3	3	3	2	1	1	1	1	1
CO5	2	2	3	3	2	1	1	1	1	1
CO (Average) Course Average	2	2.4	3	2.6	1.8	0.4	1	1.2	1.4	0.8

3-High, 2-Medium, 1-Low

UNIT-1: *Introduction:* Basic Terminology, Elementary Data Organization, Algorithm, Efficiency of an Algorithm, Time and Space Complexity, Asymptotic notations: Theta, Big-O, and Omega, Time-Space trade-off. Abstract Data Types (ADT)

UNIT-II:*Arrays*: Definition, Single and Multidimensional Arrays, Representation of Arrays: Row Major Order, and Column Major Order, Application of arrays, Sparse Matrices and their representations.

Linked Lists: Array Implementation and Dynamic Implementation of Singly Linked Lists, Doubly Linked List, Circularly Linked List, Operations on a Linked List. Insertion, Deletion, Traversal, Polynomial Representation and Addition, Generalized Linked List

Stacks: Abstract Data Type, Primitive Stack operations: Push & Pop, Array and Linked Implementation of Stack in C, Application of stack: Prefix and Postfix Expressions, Evaluation of postfix expression, Recursion, Tower of Hanoi Problem, Simulating Recursion, Principles of recursion, Tail recursion, Removal of recursion

Queues: Abstract Data Type, Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, Array and linked implementation of queues in C, Deque and Priority Queue.

Unit III:*Trees:* Basic terminology, k-ary trees, Binary Trees, Binary Tree Representation: Array Representation and Dynamic Representation, Complete Binary Tree, Algebraic Expressions, Extended Binary Trees, Array and Linked Representation of Binary trees, Tree Traversal algorithms: In order, Preorder and Post order, Binary Search Trees, Threaded Binary trees, Traversing Threaded Binary trees, Forest, Huffman algorithm, Heap, B/B+ Tree, AVL tree

Unit IV: *Searching*& Sorting:

Sequential search, Binary Search, Comparison and Analysis

Internal Sorting: Bubble Sort, Selection Sort, Insertion Sort, Two Way Merge Sort, Heap Sort, Quick Sort Hashing

UNIT V: *Graphs:* Terminology, Sequential and linked Representations of Graphs: Adjacency Matrices, Adjacency List, Adjacency Multi list, Graph Traversal: Depth First Search and Breadth First Search, Connected Component, Spanning Trees, Minimum Cost Spanning Trees: Prims and Kruskal algorithm. Shortest Path algorithm: Dijikstra Algorithm

Text Books:

1. Aaron M. Tenenbaum, YedidyahLangsam and Moshe J. Augenstein “Data Structures Using C and C++”, PHI

Reference Books:

1. Horowitz and Sahani, “Fundamentals of Data Structures”, Galgotia Publication
2. Donald Knuth, “The Art of Computer Programming”, vol. 1 and vol. 3.
3. Jean Paul Trembley and Paul G. Sorenson, “An Introduction to Data Structures with applications”, McGraw Hill
4. R. Kruse etal, “Data Structures and Program Design in C”, Pearson Education
   Lipschutz, “Data Structures” Schaum’s Outline Series, TMH